

Nonlinear Stieltjes Transport: Standalone Integration Insert

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1 Nonlinear coordinate transport of Stieltjes finite parts

All finite parts in this section use a cutoff in the named coordinate and logarithmic scale one. Thus $\text{FP} \int_0^1 x^{-1} \log^n x \, dx = 0$, while for $k \geq 2$ its value with x^{-k} is $-n!/(k-1)^{n+1}$. A density pullback is defined by $\langle \phi^* T, h \rangle = \langle T, h \circ \phi^{-1} \rangle$; its ordinary coefficient is $\phi' f(\phi)$.

Theorem 1.1 (Complete local coordinate correction). *Let ϕ be a smooth increasing endpoint-preserving diffeomorphism of $[0, 1]$, with $\lambda = \phi'(0) > 0$. Put $r(y) = \phi(y)/y$ and $L(y) = \log r(y)$. For $k \geq 1, n \geq 0$,*

$$\begin{aligned} \mathcal{A}_\phi^{k,n} &:= \phi^* \text{FP}_x(x^{-k} \log^n x \, dx) - \text{FP}_y(\phi' \phi^{-k} \log^n \phi \, dy), \\ \langle \mathcal{A}_\phi^{k,n}, h \rangle &= \frac{1}{n+1} [y^{k-1}] \phi'(y) r(y)^{-k} L(y)^{n+1} h(y). \end{aligned}$$

In particular the coefficient of $\delta^{(j)}$ is

$$\frac{(-1)^j}{(n+1)j!} [y^{k-1-j}] (\phi' r^{-k} L^{n+1}), \quad 0 \leq j \leq k-1.$$

The formula extends linearly to finite logarithmic singular expansions modulo locally integrable remainders. Its discrepancy obeys

$$\mathcal{A}_{\phi \circ \psi}(f \, dx) = \psi^* \mathcal{A}_\phi(f \, dx) + \mathcal{A}_\psi(\phi^*(f \, dx)).$$

Proof. The meromorphic distribution y^{-k-u} has residue $-h^{(k-1)}(0)/(k-1)!$ on a test function h . Taylor subtraction through degree $k-1$ leaves an integral holomorphic for $|\Re u| < 1$; each fixed spectral derivative is dominated by an integrable power times finitely many logarithms. Its regular coefficients are $(-1)^n \text{FP}(y^{-k} \log^n y)/n!$. Under $x = \phi(y)$ the full numerator is $A(u, y) = \phi' r^{-k} e^{-uL}$. The regular coefficients give the coordinate finite part of the composed logarithmic singularity. The additional coefficient comes from $-[y^{k-1}](A(u, y)h(y))/u$ and is $[y^{k-1}](\phi' r^{-k} L^{n+1} h)/(n+1)$ after multiplying by $(-1)^n n!$. This proves the action formula and the delta coefficients. Integrable remainders obey ordinary substitution. The composition rule follows by adding and subtracting $\psi^* \text{FP}(\phi^*(f \, dx))$. $\square$

An existing comparison $K = \text{FP}(f \, dx) + \kappa \delta^{(d)}$ consequently transports to

$$\phi^* K = \text{FP}(\phi' f(\phi) \, dy) + \mathcal{A}_\phi(f \, dx) + \kappa \phi^* \delta^{(d)}.$$

For $d \geq 1$ the last term has the explicit derivative expansion

$$\phi^* \delta^{(d)} = \sum_{j=1}^d (-1)^{d+j} \frac{(d-1)!}{(j-1)!} [y^{d-j}] \left(\frac{y}{\phi(y)} \right)^d \delta^{(j)}.$$

Indeed, writing $\tau = \phi^{-1}$, its action is $(-1)^d d! [x^d] h(\tau(x))$; Lagrange inversion yields the displayed coefficients. Also $\phi^* \delta = \delta$. For a second endpoint use $1 - \phi(1 - y)$ and reflect its delta derivatives with sign $(-1)^j$. This is transport of the specified distribution, not a claim that nonlinear pullback preserves circular convolution.

1.1 Harmonic Stieltjes specialization

Use

$$\zeta(1+u, x) = \frac{1}{u} + \sum_{m \geq 0} \frac{(-1)^m}{m!} \gamma_m(x) u^m, \quad \gamma_0 = -\psi.$$

Let $P_p(u) = \prod_{j=1}^p (1+u/j) = \sum_j e_{p,j} u^j$, with $P_0 = 1$ and $e_{p,j} = 0$ outside $0 \leq j \leq p$. Its coefficients are finite harmonic polynomials because $\log P_p(u) = \sum_{r \geq 1} (-1)^{r-1} H_p^{(r)} u^r / r$. Define

$$\mathcal{T}_{m,p}(L) = m! \sum_{j=0}^{\min(m,p)} \frac{(-1)^j e_{p,j} L^{m-j+1}}{(m-j+1)!}.$$

Theorem 1.2 (All-index Stieltjes correction). *For every $m, p \geq 0$,*

$$\langle \mathcal{A}_\phi(\gamma_m^{(p)} dx), h \rangle = (-1)^p p! [y^p] \phi' r^{-p-1} \mathcal{T}_{m,p}(\log r) h.$$

If $\phi(y) = y + ay^2 + O(y^3)$ and $p \geq 1$, this correction vanishes for $m \geq 2p$, while at the last possible index it equals

$$\mathcal{A}_\phi(\gamma_{2p-1}^{(p)} dx) = \frac{(2p-1)!}{p!} a^p \delta.$$

In particular $\mathcal{A}_\phi(\psi' dx) = a\delta$.

Proof. The Hurwitz shift gives the exact singularity

$$\gamma_m^{(p)}(x) = (-1)^p p! \sum_{j=0}^{\min(m,p)} \frac{(-1)^j m! e_{p,j} \log^{m-j} x}{(m-j)! x^{p+1}} + \gamma_m^{(p)}(1+x).$$

Apply Theorem 1.1 term by term. When the slope is one, $\log r = ay + O(y^2)$. For $m \geq 2p$ the smallest logarithmic power in $\mathcal{T}_{m,p}$ is at least $p+1$, so its y^p coefficient vanishes. At $m = 2p-1$ only $j = p$ survives; use $e_{p,p} = 1/p!$. $\square$

1.2 Nonlinear circle averages

Let $\chi_\lambda : [0, 1] \rightarrow [0, 1]$ be the continuous increasing lift of $\tan(\pi\chi_\lambda(y)) = \lambda \tan(\pi y)$, where $\lambda > 0$. It equals $\pi^{-1} \arctan(\lambda \tan \pi y)$ below $1/2$, equals $1/2$ there, and requires adding one to that expression above $1/2$. Put $q = (1-\lambda)/(1+\lambda)$ and $R_\lambda(y) = \chi_\lambda(y)/y$. The inverse density is the Poisson kernel

$$(\chi_\lambda^{-1})'(x) = \Pi_q(x) = \frac{1-q^2}{1-2q \cos(2\pi x) + q^2}.$$

The Taylor series of R_λ is even.

For spectral coefficient extraction set

$$F_p(u; q) = 2(2\pi)^p \Gamma(1-u) (2\pi)^u \cos \frac{\pi(p+u)}{2} \text{Li}_{-p-u}(q).$$

Theorem 1.3 (Exact nonlinear Stieltjes averages). *For $m, p \geq 0$,*

$$\begin{aligned} \text{FP} \int_0^1 \gamma_m^{(p)}(\chi_\lambda(y)) dy &= (-1)^{m+1} m! [u^{m+1}] (F_p(u; q) - P_p(u) F_p(0; q)) \\ &\quad - (-1)^p p! [y^p] R_\lambda^{-p-1} \mathcal{T}_{m,p}(\log R_\lambda). \end{aligned}$$

The last line vanishes for odd p . At Stieltjes index zero,

$$\text{FP} \int_0^1 \psi^{(2j+1)}(\chi_\lambda(y)) dy = (-1)^{j+1} \pi (2\pi)^{2j+1} \text{Li}_{-2j-1}(q).$$

These last values are rational functions of q times the displayed power of π .

Proof. The canonical periodic Stieltjes generator has nonzero Fourier coefficients

$$(2\pi in)^p \left[\Gamma(-u)(2\pi in)^u + \frac{P_p(u)}{u} \right]$$

and zero mean. This follows from the Hurwitz Fourier formula and full endpoint-residue subtraction; it is the previously established fixed-coordinate identity. Pairing it with Π_q gives $-(F_p(u; q) - P_p(u)F_p(0; q))/u$. Exponential Fourier decay justifies every fixed spectral derivative. To change to the unweighted composition average, apply Theorem 1.2 with $\phi = \chi_\lambda$ and test function $1/\chi'_\lambda$. This cancels the Jacobian in the contact coefficient. Evenness of R_λ gives the odd-order zero; differentiating the spectral factor at $u = 0$ and using $\gamma_0 = -\psi$ proves the polygamma formula. $\square$

Ordinary convergent specializations, with $\text{Li}_s^{[1]}(q) = \partial_s \text{Li}_s(q)$, are

$$\begin{aligned} \int_0^1 \left[\psi(\chi_2(y)) + \frac{1}{2y} \right] dy &= -2 \text{Li}_0^{[1]}(-1/3) - \frac{\gamma + \log \pi}{2}, \\ \int_0^1 \left[\psi'(\chi_2(y)) - \frac{1}{4y^2} \right] dy &= \frac{1}{4} + \frac{3\pi^2}{8}, \\ \int_0^1 \log \Gamma(\chi_\lambda(y)) dy &= \frac{1}{2} \log \frac{\pi(\lambda+1)}{\lambda}. \end{aligned}$$

The first two follow by subtracting their complete singular heads and using the monomial finite parts stated above. The last follows independently from Gamma reflection, symmetry of the map and the Poisson average of $\log \sin(\pi x)$. The full article supplies the all-index ordinary subtraction procedure, normalized antiderivative hierarchy, source attribution and verification artifacts. These statements do not resolve unrestricted multiple collision restrictions or the remaining Gaussian S_6 reduction.